Domestic Service Robot: Basic Coffee Preparation

Assignment D2-V1   Classical PDDL and PDDL+

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Problem and solution

The problem

- A fixed manipulator must prepare a cup of coffee in a kitchen
- Required sequence:
  - obtain the empty cup
  - fill it with water
  - add coffee powder
  - brew with the coffee machine
- Constraints:
  - static kitchen, fixed object positions
  - one object handled at a time
- Q1: classical PDDL, two instances with valid plans
- Q2: PDDL+, cooling process, cold event, timing

The solution

- Cup contents modelled as a chain of states; recipe order comes from action preconditions
- Tap, dispenser and machine are location capabilities; only the cup is grasped
- grasp and place are explicit actions on a one-slot gripper
- PDDL+: temperature fluent, autonomous cooling process, event that makes cold coffee unusable
- Checked with a real planner:
  - Q1 with pyperplan, plans of length 3 and 9
  - Q2 with ENHSP, fast delivery solved, slow delivery unsolvable

Q1: classical PDDL model

Ordering comes from preconditions, not from a script

Object states form one chain

```
(empty ?c)  ->  (has-water ?c)  ->  (has-powder ?c)  ->  (coffee-ready ?c)
```

- Stations are capabilities of locations: is-water-source, is-powder-source, is-machine. Only the cup is grasped.
- grasp and place are explicit; every station operation needs the cup placed there with the gripper free

How preconditions force the order

```
fill-water   needs (empty ?c)        only an empty cup can be filled
add-powder   needs (has-water ?c)    powder strictly after water
brew-coffee  needs (has-water ?c) and (has-powder ?c)
```

Q1: instances and plans

Same domain, only the initial state changes the difficulty

Simple: one counter does everything

```
(fill-water  cup1 counter)
(add-powder  cup1 counter)
(brew-coffee cup1 counter)
```

pyperplan: plan length 3

Complex: separate stations, cup on a shelf

```
(grasp       cup1 shelf)
(place       cup1 water-source)
(fill-water  cup1 water-source)
(grasp       cup1 water-source)
(place       cup1 powder-dispenser)
(add-powder  cup1 powder-dispenser)
(grasp       cup1 powder-dispenser)
(place       cup1 coffee-machine)
(brew-coffee cup1 coffee-machine)
```

pyperplan: plan length 9

- Planning difficulty is in the structure of the instance, not in the actions

Q2: PDDL+ model

The world changes even when the robot does nothing

Process: continuous cooling

```
(:process cooling
  :precondition (coffee-ready ?c)
  :effect (decrease
    (temperature ?c) (* #t 1)))
```

Event: fires by itself at the threshold

```
(:event coffee-got-cold
  :precondition (and (coffee-ready ?c)
    (<= (temperature ?c)
        (drinkable-temp)))
  :effect (and (not (coffee-ready ?c))
               (spoiled ?c)))
```

- Brewing and delivery take time. ENHSP has no durative actions, so each becomes start action, clock process and completion event.
- Delivery completes only if the coffee is still hot when the hand-over ends

```
start-brew -> brewing-process (clock) -> brew-done at t = 5: temperature := 90
```

Q2: timing decides plan validity

Two instances, identical except one number: delivery-time

delivery-time = 10 -> Problem Solved

```
 0.0  fill-water, add-powder,
      start-brew
 5.0  [brew-done: 90]
      grasp, start-delivery
15.0  [delivery-complete ~80]
      goal (served cup1) reached
```

delivery-time = 70 -> Problem unsolvable

```
hot window: 90 -> 40 at 1 per unit
          = 50 units  <  70 needed
coffee-got-cold fires in transit,
delivery-complete can never trigger

ENHSP: "Problem unsolvable"
```

- The planner even found a loophole in the first model: it put the cup back in the machine mid-delivery and re-brewed the spoiled coffee
    - fix: place forbidden while delivering, spoiling made irreversible. A model must encode all its invariants.

Reflection

Main challenges

- ENHSP has no durative actions, so brewing and delivery had to be rebuilt as a start action plus a clock process plus a completion event
- the planner found a loophole (re-brewing spoiled coffee mid-delivery); closing it needed extra preconditions
- ENHSP quirks: the name at is reserved, and it discretises time with a 1-second step

What I would do differently

- use exponential cooling instead of a constant rate, closer to real physics
- model water and powder as numeric resources and add a clean-up action for spoiled coffee

Limitations of the model

- states are causal milestones, not quantities: no partial fills and no real temperature curve
- no geometry, grasp feasibility or sensing (the task-motion gap); the symbolic plan assumes the physical actions just work